



ST BENEDICT'S

SUBJECT	PHYSICAL SCIENCE PAPER 1
GRADE	11
EXAMINER	Mrs Govender
MODERATOR	Mr McCoy
DATE	18 July 2024
MARKS	165
DURATION	2,5 hours

%

LEARNER NAME:

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 24 pages and a separate data sheet. Please check that your question paper is complete.
2. Read the questions carefully.
3. Answer **ALL** the questions on the question paper and hand it in at the end of the assessment. Remember to write your name and surname in the space provided above.
4. Use the data and formulae whenever necessary.
5. Show your working in all calculations.
6. Units need not be included in the working of calculations, but appropriate units should be shown in the answer.
7. Answers must be expressed in decimal format, not left as proper fractions.
8. Where appropriate, express answers to TWO decimal places.
9. It is in your own interest to write legibly and to present your work neatly.
10. Additional space is included at the end of the paper. If you run out of space for a question, use this space. Clearly indicate the question number of your answer should you use this extra space.

FOR OFFICE USE ONLY: MARKER TO ENTER MARKS

Question	1	2	3	4	5	6	7	8	9	10	Total
Mark											
Moderated Mark											
Question Total	14	20	14	20	18	11	15	14	24	12	165

Comment:

QUESTION 1**16 MARKS**

Answer these questions on the multiple-choice answer sheet below. Make a cross (X) in the box corresponding to the letter that you consider to be correct.

A	B	C	D
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Here the option C has been marked as an example.

1.1	A	B	C	D
1.2	A	B	C	D
1.3	A	B	C	D
1.4	A	B	C	D
1.5	A	B	C	D
1.6	A	B	C	D
1.7	A	B	C	D
1.8	A	B	C	D

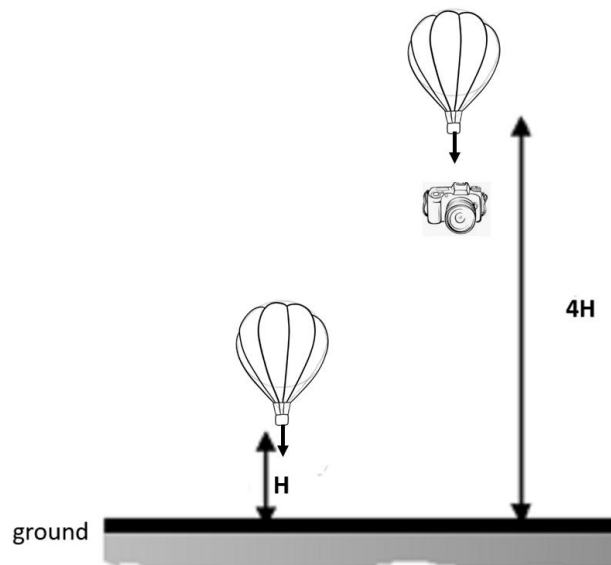
- 1.1 F_{net} can be expressed mathematically as $F_{\text{net}} = ma$
This equation consists of:

- A one scalar quantity and two vector quantities
- B two scalar quantities and one vector quantity
- C three scalar quantities
- D three vector quantities

- 1.2 A person is standing on a bathroom scale in a moving lift. Which ONE of the following motions of the lift will result in the SMALLEST reading on the scale?

- A The lift accelerates upwards
- B the lift accelerates downwards
- C The lift moves upwards at a constant velocity
- D the lift moves downwards at a constant velocity

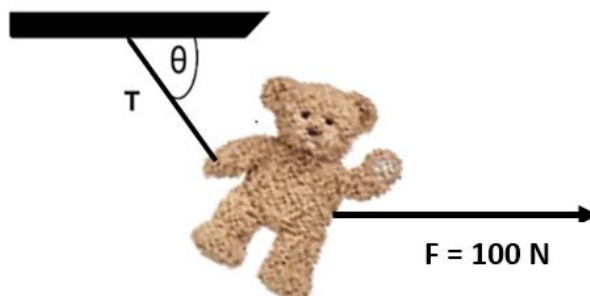
- 1.3 A camera is dropped from a hot air balloon from two different heights. In the first instance the height of the balloon from the ground is H and it takes t seconds for the camera to reach the ground. In the second instance the camera is dropped from a height of $4H$.



The time taken for the camera, dropped from the height of $4H$, to reach the ground compared to the time t of the package dropped from H is:

- A t
 - B $2t$
 - C $4t$
 - D $16t$
- 1.4 Two objects: P of mass 6 kg and Q of mass 24 kg
- A P has more inertia than Q
 - B Q has more inertia than P
 - C P and Q have the same inertia
 - D P and Q have no inertia

- 1.5 A teddy bear is tied to a cable with tension T that makes an angle of θ with the ceiling of a bedroom. The teddy bear is held stationary by a force of 100 N which pushes it to the right.

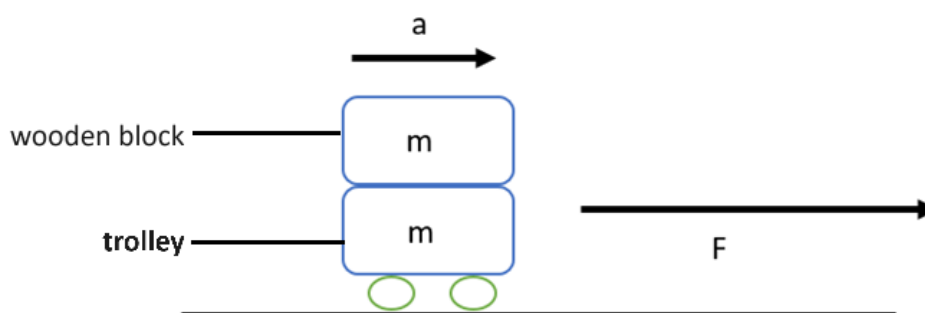


Consider these statements:

- I. $T^2 = Fg^2 + 100$
- II. $T \cos \theta = 100$
- III. $T \sin \theta = Fg$

Which of the statements are correct?

- A I and II only
 - B II and III only
 - C I and III only
 - D I, II and III
- 1.6 A constant force F is applied on a trolley of mass m with a wooden block of the SAME mass on it. The trolley-block system moves to the right with an acceleration a , on a horizontal frictionless surface.

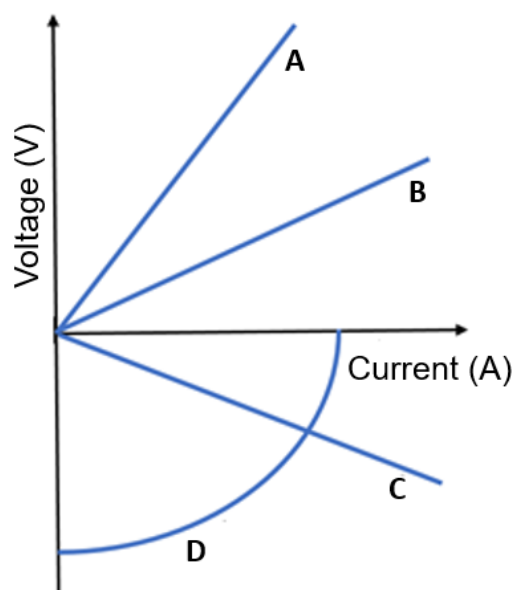


The wooden block is now removed, while still applying the SAME FORCE F .

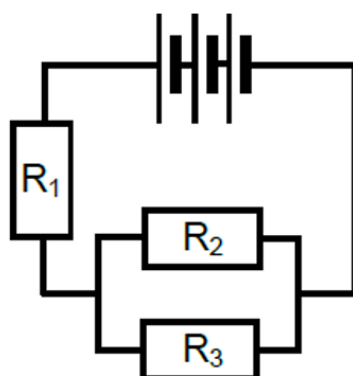
The magnitude of the acceleration without the wooden block is

- A $\frac{1}{2}a$
- B a
- C $2a$
- D $4a$

- 1.7 Which one of the following graph lines (A, B, C or D) is plotted from the results of an Ohm's law investigation where the resistor is ohmic and has the least resistance? Assume the scale on the axes of the same variable is the same across both graphs.



- 1.8 In the circuit diagram below R_1 , R_2 and R_3 are identical resistors. Ignore internal resistance.



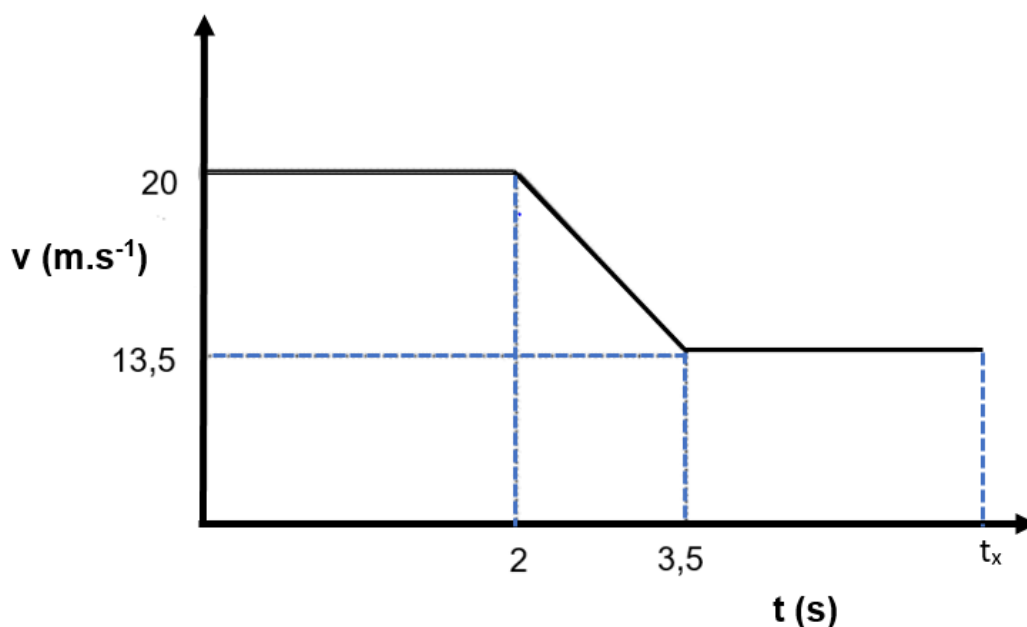
The power dissipated by R_1 is P . which one of the following is the power dissipated by R_2 ?

- A $\frac{1}{4} P$
- B $\frac{1}{2} P$
- C $2 P$
- D $4 P$

QUESTION 2 KINEMATICS – HORIZONTAL MOTION**20 MARKS**

Sam and Sikho are driving in the Kruger National Park. The speed limit in the park is $50 \text{ km}\cdot\text{h}^{-1}$ and they are travelling at $20 \text{ m}\cdot\text{s}^{-1}$ North. There is a traffic officer with a camera sitting up ahead and as soon as Sam notices the camera he slams on the brakes.

The graph below shows the velocity vs. time graph for the car from the moment the driver spots the camera.



2.1 Define *acceleration*. (1)

2.2 Use the graph to determine the acceleration of the car whilst braking. (3)

2.3 Did the driver slow down to below the required speed limit?
Show all working out. (3)

2.4 Determine the average speed of the car during the first 3,5 s of the graph. (3)

2.5 The traffic cop saw that the vehicle was speeding.

- It took the traffic cop 2,5 s after the 3,5 s indicated on the graph, to get onto his motorcycle and chase after the car at an acceleration of $0,75 \text{ m}\cdot\text{s}^{-2}$.
- The driver didn't notice the traffic cop coming up behind them and continued at a constant velocity of $13,5 \text{ m}\cdot\text{s}^{-1}$ North.

2.5.1 How long did it take him to catch up with the car from when he saw the vehicle, to time t_x . (5)

2.5.2 Sketch a displacement vs. time graph for the motion of the car for $0 - t_x$. (5)

QUESTION 3 KINEMATICS**14 MARKS**

- 3.1 A toy rocket of mass 0,45 kg is placed vertically in a bottle and ignited. It accelerates upwards at $7,5 \text{ m.s}^{-2}$. It experiences air resistance of 0,3 N for the entire flight.

- 3.1.1 Define *weight*. (2)

- 3.1.2 Describe the reaction force to the weight. (2)

- 3.1.3 Calculate the upwards force needed to accelerate the rocket vertically at $7,5 \text{ m.s}^{-2}$. (4)

- 3.1.4 Calculate how far the rocket travels in 3 seconds. (2)

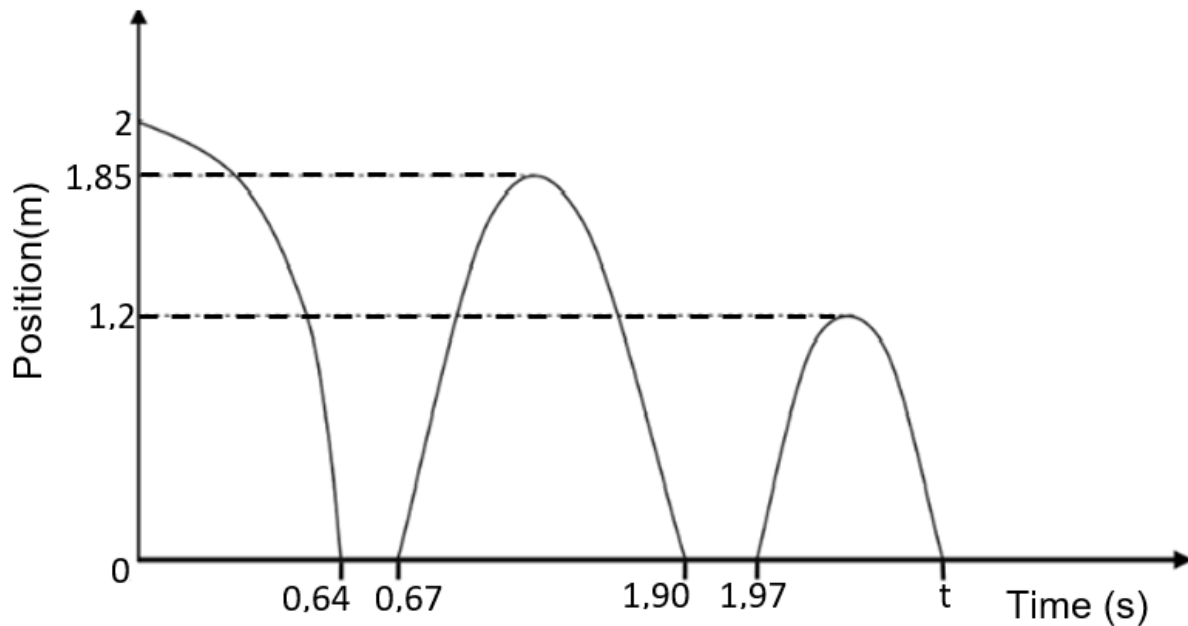
3.2 After 3 seconds of flight a 120 g piece of the nose-cone falls off and as it falls it experiences a 0,1 N frictional force.

3.2.1 Calculate the velocity of the nose-cone as it falls off. (2)

3.2.2 Calculate the net force acting on the nose-cone when it is moving down. (2)

QUESTION 4 BOUNCING BALLS**19 MARKS**

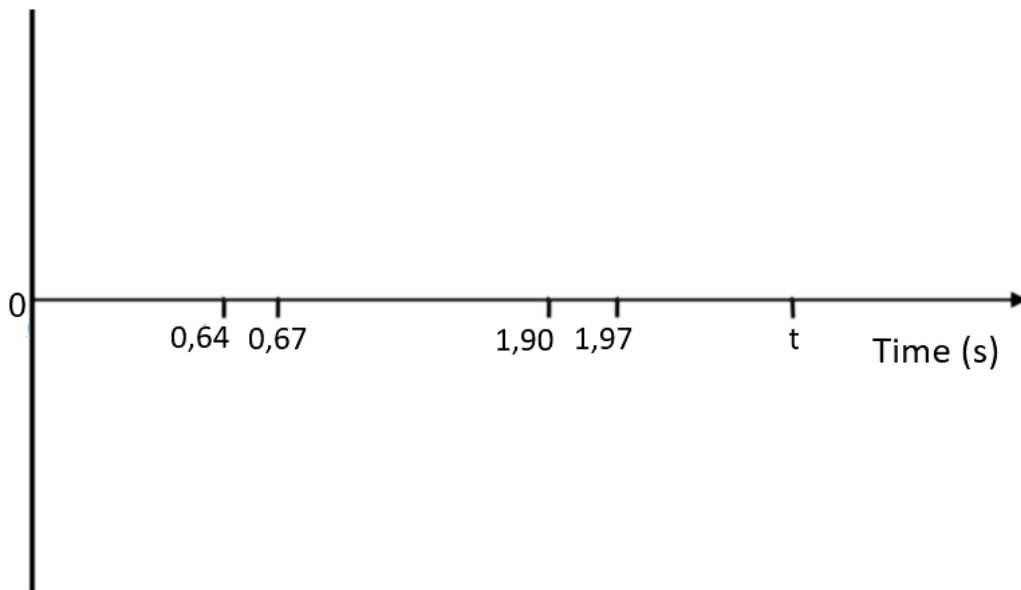
Praise drops a ball from rest and it bounces several times as shown in the position-time graph below. The mass of the ball is 40 g.



4.1 Calculate the velocity with which the ball hits the ground the first time. (3)

4.2 Calculate the magnitude of the velocity with which the ball leaves the ground at 0,67 seconds. (4)

- 4.3 Sketch the corresponding velocity-time graph for the full motion of the ball, indicating the relevant velocities you have calculated and taking upwards to be positive. (5)



- 4.4 Use your graph to calculate the magnitude of the acceleration of the ball when it was in contact with the court. (3)

- 4.5 Hence calculate the magnitude and direction of the force applied by the court on the ball. (4)

QUESTION 5 SKILLS**20 MARKS**

While on the moon, astronauts dropped a golf ball from rest and measured the time taken for the ball to reach specific heights. The results that the astronauts obtained are recorded in the table below.

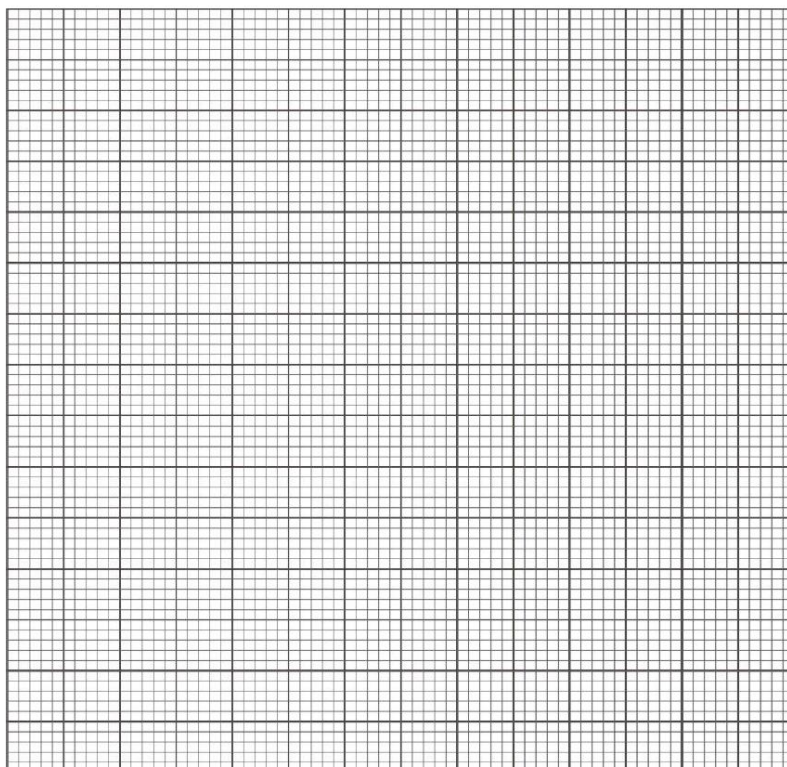
Height (m)	Time (s)	Time squared (s ²)
1,2	0,32	0,10
1,1	0,52	0,27
0,8	0,77	0,59
0,6	0,94	0,88
0,3	1,10	1,21
0,2	1,17	5.1

5.1 Fill in the missing value on the table. (2)

5.2 Explain the term '*free fall*'. (2)

5.3 Plot a graph of height (on y-axis) versus time² (on x-axis) on the graph paper. (6)

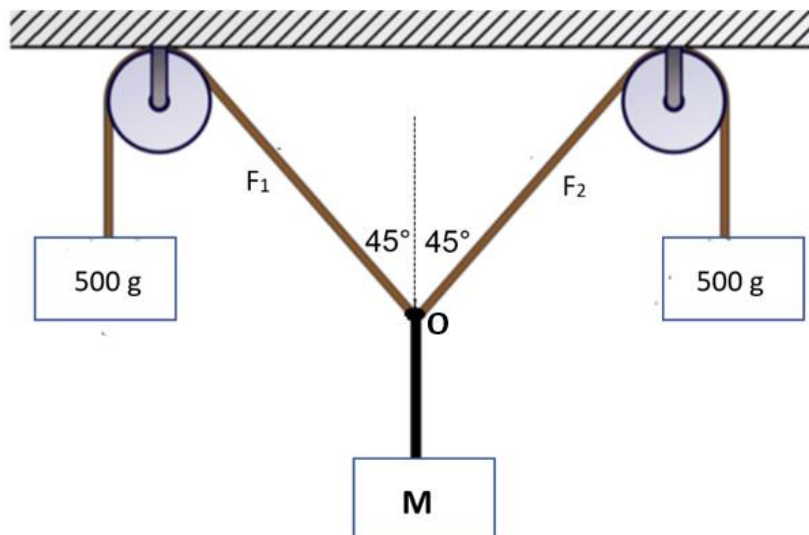
No heading is needed.



- 5.4 Calculate the gradient of your graph. (4)
- Show the values you used on your graph.
 - Include an appropriate unit in your answer.
- 5.5 Use an equation of motion and your gradient calculated in 5.4 to determine the acceleration of the golf ball on the moon. (4)
- 5.6 Without doing any further calculations, use your graph to determine the height from which the golf ball was dropped. (2)

QUESTION 6 VECTORS**11 MARKS**

In an experiment to determine the unknown mass of an object, two 500 g masses and an object of mass, M , are suspended from three strings. The string is light and inextensible and run over frictionless pulleys. When the three forces acting on knot O are in equilibrium, the angles between the two strings and the vertical are 45° as shown in the diagram.



6.1 Define a *vector*. (1)

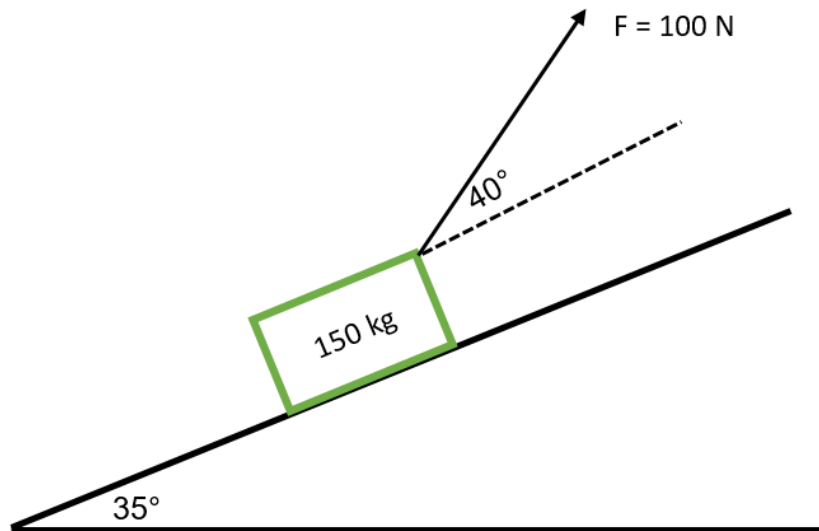
6.2 Draw a free body diagram of the forces which act on the knot at O (3)

6.3 Draw a closed vector diagram (triangle of vectors) of the forces which can be used to determine the mass of the object. (4)

6.4 Determine the mass M . (3)

QUESTION 7 FRICTION**15 MARKS**

A box of mass 150 kg is prevented from sliding down the slope by a force, F , of 100 N acting at 40° to the rough inclined plane as shown. The plane is inclined at 35° to the horizontal and the coefficient of static friction between the box and the surface is 0,7.



7.1 Draw a labelled free-body diagram showing all the forces acting on the box. (4)

7.2 Calculate the component of weight parallel to the inclined plane. (3)

7.3 Calculate the magnitude of the frictional force that is acting on the box. (4)

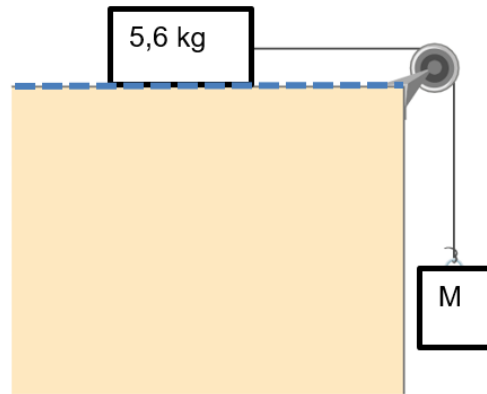
The size force **F** is slowly reduced from 100 N until the box is about to slide.

7.4 Calculate the minimum value to the force **F** for the box to remain at rest. (4)

QUESTION 8 NEWTON'S LAWS**14 MARKS**

A block of mass M is attached to a block of mass $5,6\text{ kg}$ by a light inextensible string which is passed over a frictionless pulley as show in the diagram below.

- The block of mass $5,6\text{ kg}$ is at rest on a horizontal surface with a coefficient of kinetic friction of $0,65$ and a coefficient of static friction $0,8$.
- If the hanging M was any heavier the system would not remain in equilibrium.



8.1 State Newton's second law.

(2)

8.2 Calculate the magnitude of the tension in the string while the system is in equilibrium. (3)

- 8.3 Mass (M) is increased so that the system starts to accelerate at $0,25 \text{ m.s}^{-2}$.
Determine the mass of the block which causes this acceleration. (5)

- 8.4 State *Newton's third law*. (2)

- 8.5 Identify one pair of Newton's third law forces. (2)

QUESTION 9 ELECTRIC CIRCUITS**24 MARKS**

9.1 A 12 V, 24 W kettle is connected to a supply of 12 V. The internal resistance of the supply is negligible.

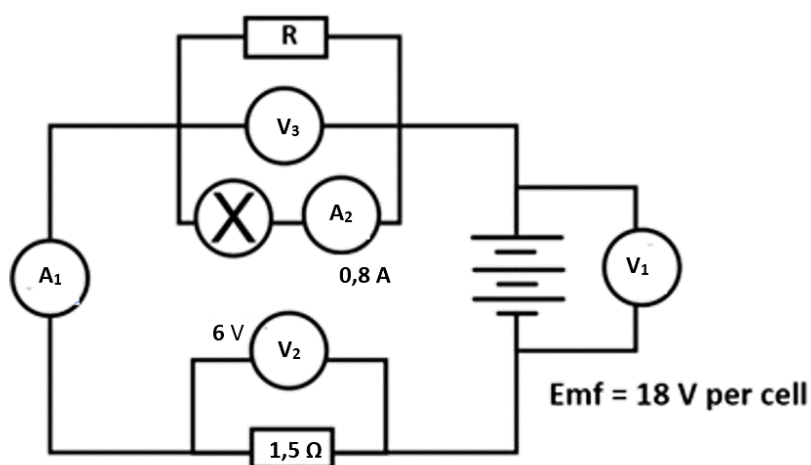
9.1.1 Define *resistance*.

(2)

9.1.2 Calculate the resistance of the kettle.

(3)

9.2 The battery in the circuit below has an emf of 18 V per cell. The reading on ammeter A_2 is 0,8 A and the voltage across the $1,5 \Omega$ resistor is 6 V. R is a resistor of unknown value and the power rating of the bulb is 40 W.



9.2.1 Explain what is meant by a power rating of “40 W”

(2)

9.2.2 Calculate the resistance of the bulb.

(3)

9.2.3 Calculate the reading on V_3 . (3)

9.2.4 Calculate the reading on ammeter A_1 . (2)

9.2.5 Calculate that resistance of resistor R . (3)

9.2.6 Define *current*. (1)

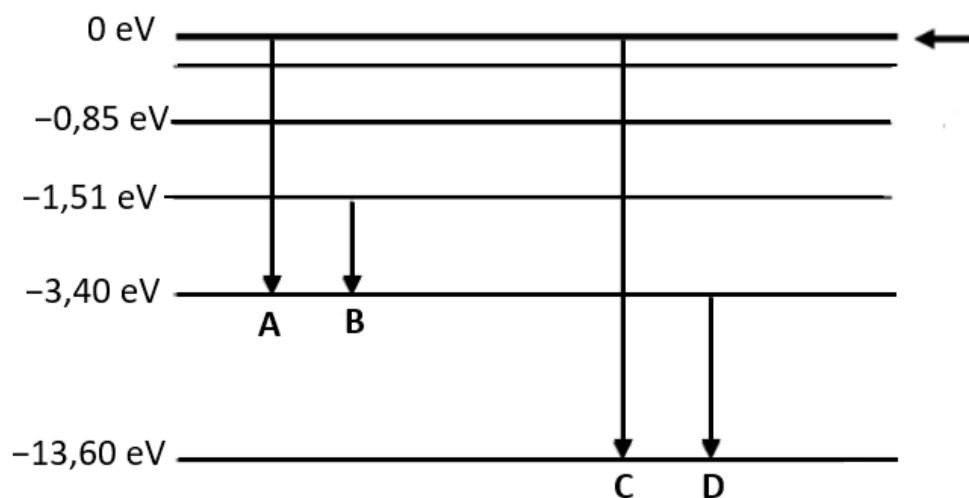
9.2.7 Resistor (R) is now removed from the circuit. How will the reading on ammeter (A_1) change?

Circle INCREASES, DECREASES or REMAINS THE SAME. (2)

9.2.8 Use a suitable equation to explain your answer in 9.2.7 (3)

QUESTION 10 WAVES**12 MARKS**

10.1 The diagram below shows four transitions (A, B, C and D) in a hydrogen atom. The diagram is not to scale.



10.1.1 Which arrow (A, B, C or D) corresponds to the greatest wavelength of light emitted? Give a reason for your answer. (3)

10.1.2 Calculate the frequency of light that would be emitted by transition D. (4)

10.2 The following diagram represents the hydrogen emission spectrum.

- The wavelengths corresponding to the violet, blue and red spectral lines are shown.
- In the diagram below and are 410 nm, 434 nm and 656 nm respectively.

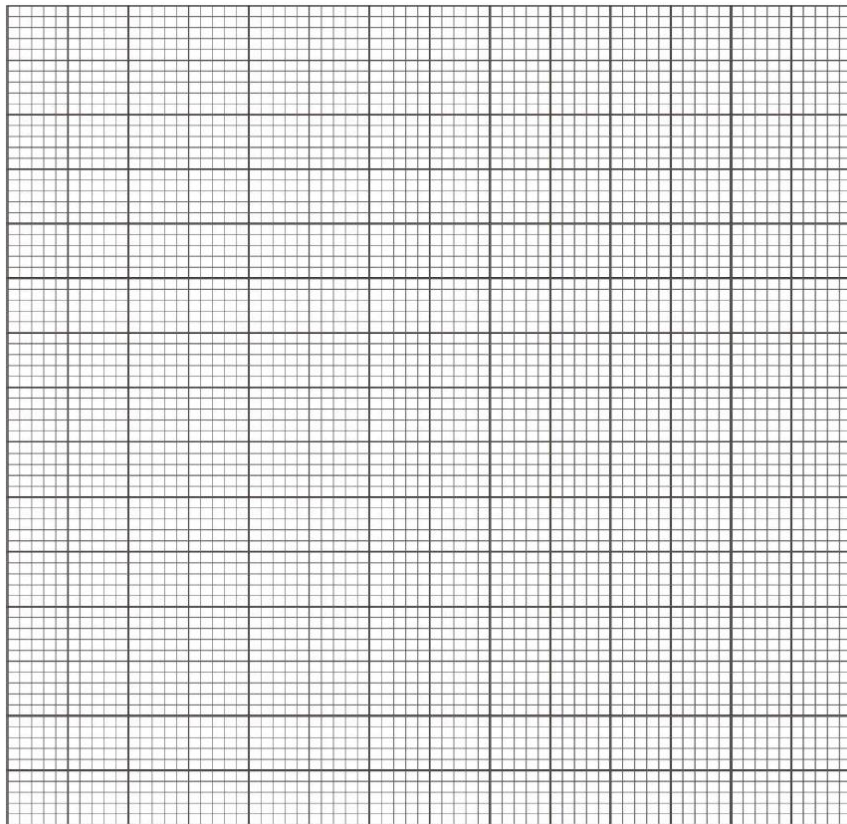


- 10.2.1 There is another line in the hydrogen emission spectrum corresponding to a colour. It has a frequency of $7,81 \times 10^{14}$ Hz. Calculate the wavelength of this spectral line. (3)

- 10.2.2 Draw an arrow on the diagram above to show the approximate position of the spectral line with frequency $7,81 \times 10^{14}$ Hz. (2)

END OF ASSESSMENT

Extra Graph Paper



ADDITIONAL SPACE

[illegible]